9th Class • Punjab Board • PCTB New Syllabus

Chapter 1: The Science of Biology

Solved notes in book order — MCQs, short questions, long questions, inquisitive questions and key points.

Chapter 1 • Units 1.1 – 1.8 • Exercise solved in book order

What this chapter is about

• Biology = the science of life. This first chapter tells us what Biology is, its main fields and branches, and how it is linked with other sciences.

• It also tells us the careers we can choose after studying Biology, what the Holy Quran says about the study of life, and why science is a team work.

• The biggest part is the scientific method: problem, observations, hypothesis, deduction, experiments and results — explained through the story of malaria.

• Every question of the textbook exercise is solved here in the same order as in the book.

A. Multiple Choice Questions

Tick the correct option. The reason after each answer helps you remember why the other options are wrong.

1. Which branch of Biology focuses on the study of the structure and function of cells?

a) Cytology
b) Microbiology
c) Histology
d) Ecology

Answer: a) Cytology — Cytology is the study of cells and their organelles; Histology deals with tissues, Microbiology with microorganisms and Ecology with organisms and their environment.

2. The study of the processes of heredity and variation in living organisms is known as:

a) Ecology
b) Genetics
c) Anatomy
d) Embryology

Answer: b) Genetics — Genetics studies the transfer of characters from parents to offspring, that is heredity and variation.

3. Insulin made through bacteria is an example of the technique of:

a) Parasitology
b) Biotechnology
c) Biochemistry
d) Histology

Answer: b) Biotechnology — Biotechnology uses living organisms or their parts (bacteria) to make useful products such as insulin for diabetic patients.

4. Heart pumps blood, stomach digests food, and kidneys excrete wastes. The statement comes from.

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a) Physiology
b) Anatomy
c) Morphology
d) Histology

Answer: a) Physiology — Physiology deals with the functioning of body parts, not with their shape (Morphology) or internal structure (Anatomy).

5. Which branch of Biology involves the study of the classification of organisms?

a) Taxonomy
b) Physiology
c) Palaeontology
d) Biogeography

Answer: a) Taxonomy — Taxonomy arranges organisms into groups on the basis of their similarities and differences.

6. Which step comes between making hypothesis and doing experiments?

a) Making deductions
b) Making observations
c) Summarizing results
d) Analysing data

Answer: a) Making deductions — The scientific method runs as observations, hypothesis, deduction, experiments, results. So deduction comes between hypothesis and experiment.

7. Which of the following is NOT a characteristic of the scientific method?

a) It relies on evidence
b) It involves formulating hypotheses
c) Hypothesis will always be correct
d) It requires rigorous testing

Answer: c) Hypothesis will always be correct — A hypothesis is only a tentative (possible) answer; experiments may prove it wrong, so it is never sure to be correct.

8. Choose the correct sequence of steps of scientific method?

a) Observations – hypothesis – deduction – experiments
b) Observations – hypothesis – law – theory
c) Hypothesis – observations – deduction – experiments
d) Law – theory – deduction – observations

Answer: a) Observations – hypothesis – deduction – experiments — First observations are made, then a hypothesis is proposed, deductions are drawn from it, and finally experiments are performed to test it.

9. People who slept near smoky fire had less chance to suffer from malaria. Why?

a) Smoke kills Plasmodium in their blood
b) Fire increases temperature and Plasmodium are killed in air
c) Mosquitoes cannot tolerate smoke and are repelled
d) Smoke kills Plasmodium present in mosquitoes

Answer: c) Mosquitoes cannot tolerate smoke and are repelled — Smoke keeps the mosquitoes away, and without the mosquito bite the Plasmodium cannot enter the body.

10. Experiments are very important in scientific method because a researcher:

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a) Always gets correct results

b) Disproves many hypotheses and gets some hypothesis proved

c) Is sure that he will prove the hypotheses

d) Gets a chance to work in the laboratory

Answer: b) Disproves many hypotheses and gets some hypothesis proved — An experiment tests several hypotheses; the wrong ones are rejected and only the proved one is accepted.

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B. Write Short Answers

1. Define the following branches of Biology: Genetics, Anatomy, Palaeontology, Marine Biology, Pathology.

Genetics: the branch of Biology that deals with the study of transfer of characters from parents to offspring. It also studies the causes of genetic disorders and helps in developing better varieties of plants and animals.

Anatomy: the branch of Biology that explores the internal physical structure of organisms, particularly of humans. It helps in disease diagnosis, in making medical devices and in improving the quality of life, e.g. study of the organs of the digestive system.

Palaeontology: the branch of Biology that deals with the study of fossils. Fossils are the remains of plants and animals preserved in rocks; their study tells us the evolutionary history of organisms, e.g. dinosaur fossils.

Marine Biology: the branch of Biology that deals with the study of life in oceans. It helps us to understand ocean biodiversity, discover new species and solve marine conservation problems, e.g. coral reefs support a wide variety of marine life.

Pathology: the study of diseases, their causes and effects. It helps in the diagnosis, prevention and treatment of diseases, e.g. a pathologist studies how uncontrolled division and spread of cells causes cancer.

2. Which branch of Biology involves the study of the development of organisms from fertilization to birth or hatching?

Answer: Embryology

Embryology is the branch of Biology that deals with the development of an organism from the fertilized egg. In this branch scientists study the formation of tissues and organs, identify birth defects and develop medical treatments.

3. How is the profession of medicine and surgery different from animal husbandry?

Medicine and Surgery	Animal Husbandry
It deals with the diagnosis and treatment of diseases of human beings.	It deals with the breeding and caring of livestock (farm animals) to improve their quality and productivity.
In surgery, the defective parts of the human body are repaired, replaced or removed.	It looks after the feeding, health and production of animals such as cows, buffaloes, goats and sheep.
It needs a 5-year Bachelor of Medicine, Bachelor of Surgery (MBBS) degree.	It needs a 4-year BS degree in Animal Husbandry.

4. Differentiate between Morphology and Physiology.

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Morphology	Physiology
It is the study of the form and structure of organisms.	It is the study of the functioning of body parts.
It studies outward appearance such as shape, colour and pattern as well as internal structures like organs.	It studies how body parts work, e.g. how the blood circulatory system transports substances in the body.
Example: study of the shape and parts of a flower or of the organs of a frog.	Example: study of how the heart pumps blood and the stomach digests food.
It tells us what a structure looks like.	It tells us what a structure does and how it works.

5. What is Computational Biology?

Computational Biology is the branch in which scientists use mathematical models, algorithms and computer simulations to understand biological systems and their relationships. It involves the analysis of biological data such as the sequence of amino acids in a protein. In this way difficult and huge biological data can be studied quickly and accurately with the help of computers.

6. What is the role of observation and experimentation in the scientific method?

Role of observation: observation is the second step of the scientific method. A scientist observes the problem with the help of his five senses and also studies the previous research on that problem. Observations may be qualitative (colour, texture) or quantitative (number of birds in a tree). Quantitative observations are more accurate because they can be measured and recorded in numbers. Good observations lead to a correct hypothesis.

Role of experimentation: experiments are the most basic step of the scientific method. They test all the hypotheses made for a problem: the wrong hypotheses are rejected and the proved one is accepted. Experiments are performed on two groups — the experimental group and the control group — so that the result is reliable. In short, observation gives the scientist facts, and experimentation that proves or disproves them, i.e. it converts a guess into a proved fact.

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Remember the difference

• Qualitative observation: cannot be measured in numbers — colour and texture of a flower.

• Quantitative observation: can be measured and recorded in numbers — number of birds on a tree.

• Experimental group: the set-up on which the factor is tested. Control group: the set-up kept normal for comparison.

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C. Write Long Answers

1. Link the study of Biology with that of Physics, Chemistry, Statistics, Geography, Economics and Computer Science.

Biology is closely linked with other natural sciences. These connections help us to understand life processes, environmental interactions and the complexity of living organisms. The links are given below.

1. Biology and Physics (Biophysics): Biophysics deals with the principles of Physics that apply to biological processes. For example, the rules of lever and motion are used to understand the working of muscles, bones and joints.

2. Biology and Chemistry (Biochemistry): Biochemistry is the study of the structure and reactions of the chemical substances present in living systems. The chemical reactions of photosynthesis and respiration are examples of Biochemistry.

3. Biology and Statistics (Biostatistics): Biostatistics uses the principles of statistics to analyse and interpret data related to living organisms. It plays a key role in biological research, healthcare and public health.

4. Biology and Geography (Biogeography): Biogeography deals with the distribution of living organisms in different geographical regions of the world. It also studies the effect of climate change on the distribution of organisms.

5. Biology and Economics (Bio-economics): Bio-economics studies organisms from the economic point of view. Scientists calculate the cost and profit of biological projects, e.g. production of a new variety of crops.

6. Biology and Computer Science (Bioinformatics / Computational Biology): Mathematical models, algorithms and computer simulations are used to understand biological systems, and biological data such as the sequence of amino acids in a protein is analysed with computational tools.

> **Easy way to remember**
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> • Physics gives the forces, Chemistry gives the molecules, Statistics gives the numbers, Geography gives the place, Economics gives the cost and Computer Science gives the tool — Biology joins them all to explain life.

2. Explain how the study of Biology can lead to different professional studies.

After FSc with Biology, students can select further studies for many careers. Some of them are:

1. Medicine and Surgery: deals with the diagnosis and treatment of diseases; in surgery the defective parts of the body are repaired, replaced or removed. It needs a 5-year MBBS degree.

2. Dentistry: dentists specialize in oral health, diagnose and treat dental diseases and perform surgeries. It needs a 4-year BDS degree.

3. Pharmacology: pharmacologists study the effects of drugs on the human body and develop new medicines. A BS in Pharmacy or a Doctor of Pharmacy (D.Pharm)

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degree is required.

4. Physiotherapy: it restores the movement and physical function of the body that has been impaired by disease or injury, using physical exercise and modalities such as massage. It needs a 4-year BS degree in Physical Therapy.

5. Fisheries and Wildlife: fisheries and wildlife departments offer jobs to biologists after a BS and MS degree in Zoology, Fisheries or Aquaculture.

6. Agriculture: agricultural scientists improve farming practices, crop production and sustainable agriculture techniques. A 4-year BS in Agriculture is required.

7. Animal Husbandry: it involves breeding and caring of livestock to improve their quality and productivity; a 4-year BS degree in Animal Husbandry is needed.

8. Horticulture: horticulturists cultivate fruits, vegetables, flowers and ornamental plants. A 4-year BS in Horticulture is required.

9. Forestry: foresters manage and conserve forests and wildlife; a 4-year BS in Forestry is necessary.

10. Farming: professionals of farming prepare animal farms, poultry farms and fruit farms, grow crops and raise animals for food and other products. It needs a 4-year BS in Agriculture or specific farming courses.

11. Biotechnology: biotechnologists use biological processes to develop products and technologies in medicine, agriculture and other fields. A 4-year BS in Biotechnology is required.

12. Forensics: forensic scientists analyse physical evidence from crime scenes in criminal investigations. A 4-year BS in Forensic Science is needed.

More careers in Biology	Major jobs
Veterinary Medicine	Diagnosis and treatment of diseases and surgeries in animals
Environmental Science	Solving issues related to pollution and natural resources
Microbiology	Research on microorganisms to understand their impact
Genetic Counselling	Providing support to people on genetic conditions and testing
Nutrition and Dietetics	Advising on proper dietary habits to promote health
Public Health	Improving the health of communities through education, policy-making and research
Biomedical Engineering	Designing and making medical equipment to improve patient care
Bioinformatics	Analysis of biological data by using computational tools

Table: More careers in Biology (as given in the textbook)

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3. Science is a collaborative field in which scientists work together to share knowledge. Prove this statement by giving examples.

Science is a collaborative field because researchers from various disciplines work together to solve complex problems. Interdisciplinary teams solve problems more efficiently by using the strengths and expertise of each discipline, and this usually leads to quicker and long-lasting solutions. Its examples are:

1. Human Genome Project: this project aimed to sequence and map the entire human genome and was completed in 2003. Researchers from many disciplines took part in it, including molecular biology, genetics, informatics and computer science.

2. Climate change research: climate change requires the collaboration of many disciplines such as atmospheric science, ecology, economics and sociology.

3. Medical research: medical research often depends on teamwork. For example, cancer research involves oncologists (cancer consultants), biologists, biochemists, geneticists, pharmacologists and statisticians.

4. Robotics and Artificial Intelligence (AI): this field is highly interdisciplinary. It involves computer science, engineering, mathematics, neuroscience and psychology, and this teamwork has produced robotic systems, autonomous vehicles, machine learning and natural language processing.

5. Space exploration: organizations like NASA and the International Space Station (ISS) involve scientists of many fields, including astrophysics, planetary science, engineering, biology and medicine.

Hence no single scientist can solve big problems alone; sharing of knowledge and skills is the real strength of science.

4. How a hypothesis is converted to theory, law and principle?

A hypothesis is a tentative (possible) answer to a scientific problem. It is converted into a theory, a law and a principle through repeated experiments as follows:

1. Hypothesis: on the basis of observations a scientist proposes a statement that may answer the problem. For example, "Plasmodium is the cause of malaria".

2. Testing by experiments: experiments are performed to test the hypothesis. If the results prove it correct, the same hypothesis is used to make further hypotheses.

3. Theory: when experiments repeatedly prove a hypothesis, it becomes a theory. A theory is supported by extensive evidence and is repeatedly validated by many researchers, e.g. the theory of evolution explains how species change over time through natural selection.

4. Law or principle: scientists keep on testing a theory and try their best to disprove it. If a theory is proved again and again by experiments, it becomes a law or principle. A scientific law is a uniform or constant fact of nature, e.g. Hardy-Weinberg law and Mendel's laws of inheritance.

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Theory	Law / Principle
An explanation based on facts that have been repeatedly confirmed through experiments.	A uniform or constant fact of nature.
It is supported by extensive evidence and validated by many researchers.	It is proved again and again by experiments and is not disproved.
It may be modified or replaced when new evidence appears.	It is accepted as a constant fact of nature.
Example: theory of evolution by natural selection.	Example: Mendel's laws of inheritance.

Theory vs Law (a common question of this chapter)

5. What are the basic steps a scientist adopts in order to solve a scientific problem?

Scientists take specific steps for doing scientific work or research. These steps are called the scientific method; for biological research they are called the biological method. The basic steps are:

1. Recognition of a scientific problem: the scientist identifies and defines the specific issue or phenomenon that he wants to investigate. For example, a biologist notices that plants in an area are growing taller than usual and develops the problem: "What factors are responsible for the increased growth of these plants?"

2. Observation: the scientist observes the problem using his five senses and also reads the previous research on the same or a related problem. Observations may be qualitative (colour and texture of a flower) or quantitative (number of birds on a tree). Quantitative observations are more accurate because they can be measured and recorded in numbers.

3. Hypothesis: on the basis of observations the scientist develops a statement that may prove the answer of the problem. This tentative answer is called a hypothesis. A hypothesis always matches the available observations, can be tested by experiments, and there is always a way to disprove it. Many hypotheses may be made for one problem.

4. Deduction: the logical result drawn from a hypothesis is called deduction. Deductions usually follow the "if-then" pattern, e.g. hypothesis: "Leaf discoloration and stunted growth of a plant are caused by a deficiency of iron in the soil"; deduction: "If iron deficiency is causing the symptoms, then adding iron to the soil will improve the colour of leaves and promote plant growth."

5. Experiments: this is the most basic step. Experiments are performed to test all the hypotheses; the wrong hypotheses are rejected and the proved one is accepted. Scientists arrange two settings: the experimental group (on which the factor is tested) and the control group (kept normal for comparison). For example, to test the necessity of carbon dioxide for photosynthesis, one plant is given no carbon dioxide

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(experimental group) and the other is given carbon dioxide (control group).

6. Results and reporting: data are gathered from the experiments and summarized with statistical analyses and graphs. The scientist lists all the references to acknowledge the sources of information, prepares a scientific report, publishes his findings in scientific journals and books, and shares them in national and international meetings and seminars.

Flow of the scientific method

• Scientific problem → Observations → Hypothesis → Deduction → Experiments (experimental and control group) → Results → Reporting → Theory → Law / Principle.

6. Describe the work of different scientists in discovering the cause of malaria.

Malaria is a common disease in many countries including Pakistan; in human history it has killed more people than any other disease. The disease was known to physicians of ancient times (more than 2000 years ago), and quinine was the only remedy for malaria from the 17th to the 20th century. Biologists solved this problem in steps:

1. Observations of the 19th century: malaria and marshy areas have some relation; quinine is an effective drug for treating malaria; drinking the water of marshy areas does not cause malaria; and Plasmodium was seen in the blood of malarial patients.

2. Laveran (1878): a French army physician, Laveran, did research on the cause of malaria. He took the blood of a malarial patient, examined it under the microscope and noticed some microorganisms in the blood. This microorganism was named Plasmodium.

3. Hypothesis and deduction: biologists developed the hypothesis "Plasmodium is the cause of malaria", and the deduction "If Plasmodium is the cause of malaria, then all malarial patients should have Plasmodium in their blood."

4. Experiments of biologists: they examined the blood samples of 100 malarial patients (experimental group) and 100 healthy persons (control group) under the microscope. Most of the malarial patients had Plasmodium in their blood, so the hypothesis was proved true.

5. A. F. A. King (1883): the next problem was how Plasmodium gets into the blood of man. King, a physician, listed 20 observations. Important ones were: people who slept outdoors had more chances of getting malaria than those who slept indoors; people who slept under fine mosquito nets had less chance of malaria; and individuals who slept near a smoky fire usually did not get malaria. On this basis King suggested the hypothesis: "Mosquitoes transmit Plasmodium and so are involved in the spread of malaria." The deduction was: "If mosquitoes are involved in the spread of malaria, then Plasmodium should be present in mosquitoes."

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6. Ronald Ross (in the 1880s): Ross, a British army physician working in India, performed important experiments. He allowed a female Anopheles mosquito to bite a malarial patient, killed the mosquito and found Plasmodium multiplying in its stomach. Then he allowed female Culex mosquitoes to bite a sparrow suffering from malaria and found that Plasmodium multiplied in the wall of the mosquito's stomach and moved into its salivary glands. When these infected mosquitoes bit healthy sparrows, the sparrows got malaria and many Plasmodia were found in their blood. So the hypothesis was proved: mosquitoes transmit Plasmodium.

7. Experiments on human beings (1898): in the end Italian biologists allowed an Anopheles mosquito to bite a malarial patient and then allowed the infected mosquito to bite a healthy man. The man later became ill with malaria. In this way it was confirmed that mosquitoes transmit Plasmodium and so are involved in the spread of malaria.

7. Write a descriptive note on the experiments performed by Ross.

Ronald Ross was a British army physician working in India. In the 1880s he performed important experiments to test the deduction that Plasmodium should be present in mosquitoes if mosquitoes spread malaria. His work can be described in the following steps:

1. Experiment with Anopheles mosquito: Ross allowed a female Anopheles mosquito to bite a malarial patient. He then killed this mosquito and found Plasmodium multiplying in its stomach. This showed that Plasmodium can live and multiply inside the mosquito.

2. Idea of the next experiment: he thought of allowing an infected mosquito to bite a healthy person; if the hypothesis was true, that healthy person would get malaria. But he did not use human beings for such a risky experiment.

3. Experiment with sparrows: Ross performed the experiment again, but used sparrows instead of man. He allowed female Culex mosquitoes to bite a sparrow suffering from malaria and then studied some of these mosquitoes at different times.

4. Result of the experiment: he found that Plasmodium multiplied in the walls of the mosquito's stomach and then moved into its salivary glands. He then allowed some infected mosquitoes to bite healthy sparrows; these healthy sparrows got malaria, and when he examined their blood he found many Plasmodia in it.

5. Conclusion: the result proved the hypothesis that mosquitoes transmit Plasmodium and so are involved in the spread of malaria. Ross's experiments also showed the importance of the control and experimental groups, because healthy sparrows were kept for comparison with the infected ones.

D. Inquisitive Questions

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1. Why is it important to classify biology into different branches such as botany, zoology, and microbiology? How does specialization benefit scientific research?

Biology is a vast field that explores the incredible diversity of life on Earth. To understand this complexity, scientists have divided Biology into three main fields — Zoology (study of animals), Botany (study of plants) and Microbiology (study of microorganisms) — and then into many branches such as Cytology, Genetics, Ecology and Physiology.

Importance of classification into branches: (i) the study of life becomes easy and organized, because each branch deals with one aspect of life; (ii) a scientist can go deeply into one field instead of learning everything superficially; (iii) it becomes easy to identify new species and to study evolutionary relationships; (iv) knowledge of one branch can be used in another branch, e.g. Genetics is used in Agriculture to develop better varieties of crops.

Benefit of specialization to scientific research: a specialist knows the techniques, instruments and literature of his own field very well. So he works faster, his results are more accurate and reliable, and he can solve difficult problems of that field. Specialists of different branches then work together as a team (as in the Human Genome Project and in cancer research), and this collaboration gives quicker and long-lasting solutions. In short, branches divide the work and specialization makes each part of that work perfect.

2. How can a scientist apply the scientific method to confirm an observation that a certain plant species grows more quickly in shady places than in direct sunlight?

The scientist will apply the steps of the scientific method in the following way.

1. Recognition of the problem: he defines the problem as "Does the plant species grow more quickly in shady places than in direct sunlight?"

2. Observation: he observes plants of that species at both places and records quantitative data, such as the height of 20 plants grown in shade and 20 plants grown in direct sunlight, measured after the same number of days.

3. Hypothesis: on the basis of these observations he proposes the hypothesis: "Shade (less light) increases the growth rate of this plant species."

4. Deduction: he draws an "if-then" deduction: "If shade increases the growth of this plant, then plants kept in shade will grow taller than the plants kept in direct sunlight under the same conditions of water, soil and temperature."

5. Experiments: he takes two similar sets of plants of the same age and same soil. One set is placed in shade (experimental group) and the other in direct sunlight (control group). Water, soil and temperature are kept the same for both sets so that only light is changed.

6. Results: after a fixed time he measures the height of the plants of both sets and compares them with the help of graphs and statistical analysis. If the shaded plants are clearly taller, the hypothesis is supported; if not, it is rejected and a new hypothesis is made.

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7. Reporting: he writes a scientific report with his data and references, publishes it in a journal and shares it with other scientists. If the result is proved again and again by other researchers, it may become a theory and later a law or principle.

E. More Important Questions (from the text)

1. What instructions does the Holy Quran give about the study of life?

The Holy Quran gives several verses that highlight the study of life and encourage us to explore and reflect on it:

1. Origin of life from water: "We made every living thing from water" (Sura Al-Anbiya, Verse 30). The Quran mentions in many verses that living things were created from water. Water is described as a divine blessing from Allah, and the average water content of different organisms ranges between 60% and 90%. This verse hints at the common origin of all living things in water.

2. Creation of man from clay: "He made man from clay like the potter" (Sura Al-Rehman, Verse 14). By the hints of these verses we can find the events that occurred in the creation of human beings; we are advised to think over the possible ways through which such events might have occurred.

3. Stages of development of the embryo: "Then fashioned We the drop a clot, then fashioned We the clot a little lump, then fashioned We the little lump bones, then clothed the bones with flesh" (Sura Al-Mominoon, Verse 14). This hint is about the method of development of animals including human beings.

4. Common origin and modification of animals: "Allah hath created every animal from water. Then some of them creep up over their bellies, others walk on two legs, and others on four" (Sura Al-Nur, Verse 45). This verse explains that Allah created early life in water (fishes) and then animals with limbs appeared — some creeping on their bellies, some walking on two legs and some on four.

So the Holy Quran not only tells us about the origin of life but also invites us to study living things scientifically.

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2. What is the difference between a hypothesis, a theory and a law?

Hypothesis	Theory	Law / Principle
A tentative (possible) answer to a scientific problem, made on the basis of observations.	A hypothesis that has been repeatedly proved by experiments and is supported by extensive evidence.	A theory that has been proved again and again by experiments and is not disproved.
It is not yet proved; it may be proved wrong.	It is validated by many researchers.	It is a uniform or constant fact of nature.
Example: "Plasmodium is the cause of malaria."	Example: theory of evolution by natural selection.	Example: Mendel's laws of inheritance, Hardy-Weinberg law.

Points to Remember

Biology is the science of life; the word comes from two Greek words, "bios" (life) and "logos" (study). It explores the structures, functions and interactions of living organisms.

Major fields of Biology: Zoology (animals), Botany (plants) and Microbiology (microorganisms such as bacteria and microscopic fungi).

Biology is linked with other sciences through Biochemistry, Biophysics, Computational Biology, Biogeography, Biostatistics, Biotechnology and Bio-economics.

Careers in Biology: medicine and surgery, dentistry, pharmacology, physiotherapy, fisheries and wildlife, agriculture, animal husbandry, horticulture, forestry, farming, biotechnology and forensics.

The Holy Quran instructs us to study life: all living things were made from water, man was made from clay, and the stages of embryonic development and the modification of animals are also described.

Science is collaborative: examples are the Human Genome Project (completed in 2003), climate change research, cancer research, robotics and AI, and space exploration (NASA, ISS).

Scientific method: recognition of a scientific problem, observation, hypothesis, deduction, experiments and results.

Hypothesis: a tentative answer to the scientific problem, which can be tested and can be disproved.

Deduction: the logical result of a hypothesis, usually written in the "if-then" form.

Experiments: they disprove wrong hypotheses and prove the correct one; they need an experimental group and a control group.

Theory: an explanation based on facts that have been repeatedly confirmed through experiments. When a theory is proved again and again, it becomes a **law or principle** — a uniform or constant fact of nature.

Malaria: caused by Plasmodium, which was first seen by Laveran (1878) in the blood of a malarial patient. A. F. A. King (1883) observed that mosquitoes are involved, and Ronald Ross proved in the 1880s that mosquitoes transmit Plasmodium. In 1898 Italian biologists confirmed it in human beings. Quinine was the only remedy for malaria from the 17th to the 20th century.

Branch	Study of
Cytology	Cells
Histology	Tissues
Morphology	Form and structure of organisms
Anatomy	Internal physical structure of organisms
Physiology	Functioning of body parts

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Branch	Study of
Embryology	Development of organism from fertilized egg
Genetics	Transfer of characters from parents to offspring
Molecular Biology	Biological molecules such as carbohydrates, proteins, lipids and nucleic acids
Palaeontology	Fossils
Taxonomy	Classification of organisms
Ecology	Relationships between organisms and their environment
Marine Biology	Life in oceans
Pathology	Diseases, their causes and effects
Immunology	Components of the immune system and their role against diseases
Pharmacology	Drugs and their effects on the body

Table: Main branches of Biology (Key Points of the chapter)

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d); there is no need to copy the full sentence.

• Short question: answer in 2 to 4 sentences. If the question says "define", give the definition and one example.

• Difference question: always draw a two-column table with proper headings.

• Long question: write a small heading, then numbered points or steps. In questions of the scientific method and malaria, the examiner looks for the correct sequence, the names with years (Laveran 1878, King 1883, Ross 1880s, Italian biologists 1898) and a closing line.